

Equations of the form $\cos(x) = b$

№1. Solve the following equations:

1. $\cos 3x = -\frac{1}{2};$

3. $\cos \frac{2\pi x}{3} = 0;$

5. $\cos \frac{5}{6}x = \sqrt{3};$

2. $\cos 6x = 1;$

4. $\cos 9x = -\frac{1}{5};$

6. $\cos\left(-\frac{x}{3}\right) = \frac{\sqrt{3}}{3}.$

№2. Solve the following equations:

1. $\cos\left(x + \frac{\pi}{6}\right) = \frac{\sqrt{2}}{2};$

3. $\cos\left(\frac{x}{6} - 2\right) = -1;$

2. $\cos\left(\frac{\pi}{4} - x\right) = -\frac{\sqrt{3}}{2};$

4. $2\cos\left(\frac{\pi}{8} - 3x\right) + 1 = 0.$

№3. Find the greatest negative root of the equation $\cos\left(x - \frac{\pi}{6}\right) = \frac{1}{2}.$

№4. How many roots of the equation $\cos 3x = \frac{\sqrt{3}}{2}$ belong to an interval $\left[-\frac{\pi}{2}; \pi\right]$?

№5. Solve the following equations:

1. $\cos \frac{2\pi}{x} = 1;$

2. $\cos \pi\sqrt{x} = -\frac{\sqrt{3}}{2};$

3. $\cos \cos x = \frac{1}{2}.$